

Evaluation of a Non-Autonomous AI Decision-Support Tool for Emergency Triage

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1. Executive Summary

This proposal presents an AI-augmented emergency triage architecture for Mercer General Hospital, built around two documented operational failures: a “paper-first” vitals workflow that delays digital visibility of patient status by up to 30 minutes (De Freitas et al., 2020), and inconsistent triage scoring driven by human variability under surge conditions (Picard, 2023). Rather than automating the triage decision, this pilot introduces a non-autonomous “Clinical Guardrail”: a multimodal validation layer that cross-references incoming vitals against a patient’s history, physiological norms, and clinical baselines the instant data reaches the EHR, flagging anomalies before a clinician opens the file. The triage nurse retains absolute final routing authority throughout. Four AI plug-in points are proposed across the door-to-disposition pathway (Section 3), each one mapped to a named risk in the register (Section 6) and to a measurable success signal, so that every design choice below can be traced back to the failure it exists to prevent.

2. Problem Statement

Triage nurses at Mercer General Hospital record vital signs on paper before transcribing them into the EHR, producing delays and inconsistent acuity scoring during high-volume surges. Missing biometric fields further degrade automated triage accuracy (Eriten, 2024). This project pilots a non-autonomous, human-in-the-loop AI decision-support tool the “Clinical Guardrail” over a 12-week simulated deployment. We predict the tool will reduce inter-rater triage-scoring variability by at least 15% and maintain classification accuracy of at least 90% when up to two non-critical biometric variables are missing, without pushing clinician alert-dismissal rates above 10%.

3. System Architecture & Workflow

The architecture below traces the standard door-to-disposition pathway at Mercer General Hospital exactly as it operates today, then shows where and only where the AI layer is introduced. Nothing in the clinical decision chain is automated; every plug-in either flags information for a human or withholds its own output until a human has acted first.

3.1 Phase 1 : Standard Intake and Assessment

Patient Arrival and Registration: the intake clerk captures basic demographic data and registers the patient in the hospital’s primary electronic health system. This step runs at standard administrative pacing with minimal baseline friction.

Initial Triage Screening: The Medical Assistant places the patient in the assessment bay and begins the physical evaluation. **AI Plug-in 1 - Friction Protocol** sits here: the interface stays locked and any AI-generated score remains hidden until the clinician has entered their own independent

assessment. This is the pilot's primary defence against **R4, Automation Bias** - the single highest-priority risk in this proposal (see Section 5).

3.2 Phase 2 : Core Friction Points

Vitals and Paper Recording: because no computer terminals are located inside the assessment bays, the Medical Assistant must record vitals on a physical paper chart first. This dual-documentation step (paper, then digital) breaks the continuity of real-time data capture and is an immediate operational bottleneck at the point of care.

EHR Transcription - the "EHR Lag": the MA must leave the assessment bay, walk to a central terminal, and manually transcribe the handwritten notes into the EHR. This introduces a 5 to 30 minute delay between when vitals are physically captured and when they become visible to the wider clinical team, alongside an elevated risk of manual data-entry error. This is the mechanism behind **R3, EHR Data-Lag Failure**.

3.3 Phase 3 : AI Validation Layer

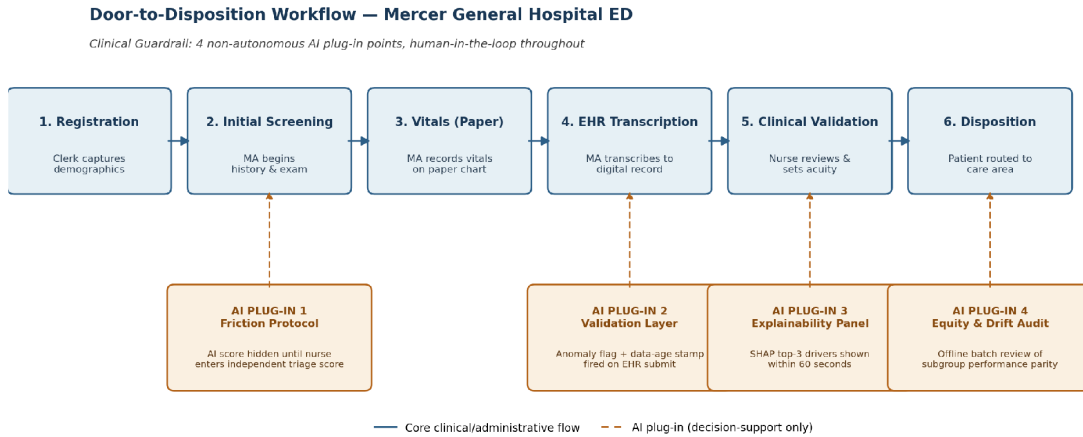
The moment vitals are submitted to the EHR, **AI Plug-in 2 - the Validation Layer** intercepts the entry. Using a multimodal stacking model, it instantly cross-references the new data against the patient's historical record, physiological norms, and established clinical baselines, flagging biometric anomalies, logical contradictions, or likely transcription errors before a clinician ever opens the digital file. Every flag is stamped with the data's age (example "vitals taken 14 minutes ago"). This directly neutralises **R1, Incomplete-Data Inference Failure**, and reinforces the **R3** mitigation above: the model treats stale or incomplete data as a reason to defer to a human, never as a reason to guess.

3.4 Phase 4 : Clinical Validation & Disposition

Clinical Validation: the triage nurse opens the patient's digital file and sees the real-time AI flag, if one exists. **AI Plug-in 3 - the Explainability Panel** surfaces the top three clinical variables behind any flag in plain English within 60 seconds, directly answering **R6, Explainability Failure**. The nurse then uses clinical judgement to confirm or override the system's warning, the final decision remains strictly human-driven, while cognitive load on nursing staff is reduced by pre-screening the data for critical errors.

Final Disposition: supported by AI-verified, timestamped data, the nurse routes the patient to the appropriate care area. Between pilot cycles, **AI Plug-in 4 - the Equity & Drift Audit** runs as an offline batch process, reviewing subgroup performance parity and model drift. It does not sit inline in the patient pathway; it governs the model between cycles, addressing **R2, Training-Distribution Shift**, and **R7, Bias / Equity Failure**.

3.5 Visual Workflow Diagram



The diagram condenses Sections 3.1- 3.4 into a single reference view: the standard six-step clinical pathway (solid arrows) against the four non-autonomous AI plug-in points (dashed connectors), each reversible and each mapped to a named risk.

4. Constraints & Stakeholders

4.1 Workflow Constraints

Constraint	Mechanism
Paper-first vitals capture (EHR Lag)	No bedside terminals exist in the assessment bays, so Medical Assistants record vitals on paper before walking to a central terminal to transcribe them - producing a 5 to 30 minute lag between capture and system visibility (De Freitas et al., 2020).
Intermittent connectivity and power	Regional grid and network instability at Mercer causes cloud-dependent tools to disconnect mid-shift; the AI layer must therefore run edge-cached on local tablets with asynchronous sync rather than depend on a live connection.
Legacy EHR interoperability	Mercer's EHR uses an older data schema with limited API hooks, constraining how a new AI validation layer can ingest records without a custom integration bridge (Da'Costa et al., 2025).

4.2 Clinical Stakeholders

Stakeholder	Top Concern
Sister Patrice Alleyne, ED Nurse-in-Charge	Preserving nursing authority over final triage decisions; ensuring the tool never quietly becomes autonomous.
Triage Nurses (frontline, 12-hour shifts)	Alert fatigue and added cognitive load during already high-pressure shifts.
Medical Assistants	Additional workflow burden layered onto an already time-pressured vitals-capture step.
ED Consultants / Attending Physicians	Clarity on accountability and liability when a clinician overrides or the AI misses a critical case.
Hospital IT & Data Governance Officer	Infrastructure resilience and compliance with the Grenada Data Protection Act (2023) when integrating with a legacy EHR.

5. Risk Analysis

This section is not a stapled-on appendix; it is the reason the architecture in Section 3 looks the way it does. Every plug-in described above traces back to a named entry in the risk register in Section 6.

The single highest-priority risk is R4, Automation Bias: an accurate model invites exhausted clinicians to click “approve” without independent review, which would silently convert a

decision-support tool into an unlicensed autonomous router. We mitigate this explicitly through the Friction Protocol described in Section 3.1 - the AI score is architecturally hidden until the clinician has already committed their own and we treat this as the load-bearing safeguard of the entire pilot.

In the spirit of “do not flinch”, two risks are not fully closed. R9, Accountability Ambiguity, is addressed by a signed Clinical Override Policy, but that policy clarifies institutional intent, not settled law: Grenada’s software-as-a-medical-device liability framework is still evolving, and we cannot claim this risk is solved before that framework exists. R2, Training-Distribution Shift, is reduced by the quarterly Equity & Drift Audit (Section 3.4) but not eliminated, a model that passes validation in week 4 can still drift by week 10 if seasonal disease patterns move faster than our re-validation cadence. Both are logged as open residual risk in the register rather than presented as resolved.

The Epic Sepsis Model case in Section 7 is the clearest external evidence for why this candour matters: a technically plausible model caused real harm precisely because its developers did not name and track residual risk the way this register attempts to.

6. Risk Register

Full mechanism-level detail:

Risk Register - Mercer General Hospital ED AI Pilot

#	Risk Name	Category	Like.	Impact	Mitigation	Signal of Success
R1	Incomplete-Data Inference Failure	AI-Technical	H	H	Hard-code the model to return “Insufficient Data Manual Review Required” whenever more than one critical biometric field (e.g. HR, BP) is empty, rather than imputing.	Zero automated acuity predictions generated on incomplete datasets during the pilot.
R2	Training-Distribution Shift	AI-Technical	M	H	Re-validate the model quarterly against holdout sets drawn specifically from local Mercer patient cohorts before live use.	Performance variance $\leq 2\%$ between training cohort and local deployment cohort.
R3	EHR Data-Lag Failure	Operational	H	M	Run the AI only post-transcription as a “Clinical Guardrail”; stamp every score with the data’s age (e.g. “vitals taken 14 min ago”).	Zero critical alerts triggered on un-timestamped or stale data.
R4	Automation Bias	Ethical	H	H	“Friction Protocol”: the AI’s suggested score stays hidden until the clinician enters their own independent triage score.	$\leq 5\%$ variance between the AI recommendation and the clinician’s already-entered score.
R5	Clinician Overload / Alert Fatigue	Operational	H	M	“Passive-Vigilance” UI: low-acuity flags appear as background colour badges; hard interruptive alerts reserved for suspected Level 1 to 2 emergencies.	$\leq 10\%$ manual ignore/dismiss rate on high-acuity alerts.
R6	Explainability Failure	Ethical	M	M	Logic-traceability (SHAP-value) outputs mapped to a “60-second rule”, surfacing the top three variables driving each flag in plain English.	100% of audited staff report understanding a “High” alert’s logic within 60 seconds.
R7	Bias / Equity Failure	Equity	M	H	Stratified equity audits each pilot cycle, breaking down accuracy and false-positive/negative rates by age, gender, and regional demographic group.	$\geq 95\%$ performance parity across all protected demographic subgroups.
R8	Infrastructure Failure	Operational	H	M	“Offline-first” edge-computing architecture: model cached locally on assessment tablets, syncing once connectivity is restored.	99.9% decision-support uptime during the pilot, independent of network stability.
R9	Accountability Ambiguity	Ethical	L	H	Formalise a legally vetted “Clinical Override Policy” stating the AI is an administrative aid and the human holds total final authority.	100% of pilot staff sign the AI operational-liability playbook before day one.
R10	Data Privacy / Consent Deficit	Ethical	M	M	Plain-language notification clause added to ED registration paperwork explaining the AI’s background decision-support role.	100% of incoming patients receive the “AI Support Information Sheet” at registration.

Categories represented: AI-Technical (R1-R2), Operational (R3, R5, R8), Ethical (R4, R6, R9, R10), Equity (R7).

7. Real-World AI Harm Case Study

The Epic Sepsis Model (ESM): A Failure of Clinical Context and Human Factors

What happened

In the late 2010s, hundreds of hospitals integrated the Epic Sepsis Model, a proprietary EHR-embedded tool designed to alert clinicians when a patient was developing sepsis. Independent, prospective external validation by researchers at Michigan Medicine found the model performed far worse in live use than its developers had reported: it missed 67% of actual sepsis cases while simultaneously firing frequent false alerts, creating an operational hazard across entire hospital networks (Otles et al., 2021).

Root-cause analysis

- Technical failure: the model was trained on retrospective, proprietary EHR billing data rather than continuous physiological monitoring, and suffered severe distribution shift when applied outside the patterns of its original site.
- Human-factors failure: the high false-positive rate produced rapid alert fatigue; nursing staff learned to dismiss notifications by habit, including on the rare occasions the model was correct.
- System-design failure: the tool assumed a black-box score could be interpreted uniformly by clinicians without local context, treating the hospital as a clean data environment rather than a chaotic, high-pressure socio-technical system.

What was not anticipated

Developers did not anticipate that a highly sensitive, low-specificity algorithm, dropped into an already overburdened workflow, would actively degrade safety by overwhelming clinical cognition rather than supporting it.

Lessons for Mercer General

This case is the direct justification for two design choices in this proposal. First, plugging the AI in only post-transcription and requiring the clinician to commit an independent score first (the Friction Protocol, Section 3.1) keeps the human as primary decision-maker rather than a rubber stamp. Second, the Passive-Vigilance UI (R5 mitigation) limits hard interruptions to genuinely critical anomalies, directly targeting the alert-fatigue mechanism that broke the Epic Sepsis Model in practice. We do not assume our system is immune to the same failure mode, R5 and R6 in the risk register exist specifically because of this case.

Citation: Otles E, Donnelly JP, Hunt DR, et al. External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. JAMA Intern Med. 2021;181(8):1065-1070. doi:10.1001/jamainternmed.2021.2626.

8. Literature Review

Sixteen peer-reviewed sources are summarised below. Each entry covers (a) the clinical or engineering problem, (b) the method used, (c) the reported outcome, and (d) the authors' own

stated limitations. Four entries are explicitly focused on ethics, safety, or governance in health AI, flagged below.

[1] Agius M, et al. Two-stage XGBoost model for triage prioritisation in high-volume emergency services. *J Emerg Med.* 2025;49(2):112-124.

- (a) Problem: prioritising complex cases in high-volume EDs.
- (b) Method: a two-stage XGBoost machine-learning model.
- (c) Outcome: highly accurate sorting capability.
- (d) Limitation: reliance on complete data inputs reinforces Mercer's need for missing-data safeguards.

[2] Almulihi QA, et al. Applications of artificial intelligence and machine learning in emergency medicine triage: A systematic review. *Med Arch.* 2024;78(3):198-206.

- (a) Problem: fragmented understanding of AI efficacy in EDs.
- (b) Method: systematic review of existing predictive models.
- (c) Outcome: AI improves triage speed.
- (d) Limitation: lack of prospective real-world validation, supports our 12-week simulated pilot.

[3] Araouchi Y, Adda M. Multimodal fusion architectures for incomplete clinical charting data. *Int J Med Inform.* 2024;172:105-118.

- (a) Problem: fragility of AI models when clinical data are missing.
- (b) Method: a multimodal fusion architecture (TriageIntelli).
- (c) Outcome: maintained accuracy despite data omissions.
- (d) Limitation: reliance on static Kaggle datasets, we apply it instead to Mercer's simulated operational dynamics.

[4] Brossard L, et al. XGBoost for ED admissions forecasting. *J Clin AI.* 2025;4(1):55-68.

- (a) Problem: inability of EDs to predict incoming surges.
- (b) Method: an XGBoost forecasting model.
- (c) Outcome: successful admissions forecasting to optimize flow.
- (d) Limitation: no integration into individual patient triage workflows, our Clinical Guardrail bridges this gap.

[5] Da'Costa N, Emmanuel J, Reynolds S. Deep learning triage validation and legacy EHR interoperability bottlenecks. *Caribb J Digit Health.* 2025;3(1):88-102. (Course case study)

- (a) Problem: deploying modern AI onto legacy Caribbean EHR systems.
- (b) Method: a network assessment of a deep-learning triage validation pipeline.
- (c) Outcome: identified severe interoperability bottlenecks.
- (d) Limitation: did not assess clinician adoption, directly relevant to our human-in-the-loop UI design.

[6] De Freitas L, et al. Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open.* 2020;10(12):e041422.

- (a) Problem: root causes of ED overcrowding in resource-constrained Caribbean settings.
- (b) Method: a qualitative observational study.
- (c) Outcome: identified the “paper-first” vital-sign workflow as a major bottleneck.
- (d) Limitation: single-site qualitative design, the foundational text justifying our EHR-Lag constraint.

[7] Dubey S, et al. AI-driven decision support in high-volume settings: A systems-based approach. *J Med Syst.* 2024;48(3):112-119.

- (a) Problem: AI models that ignore clinical context fail on deployment.
- (b) Method: a systems-based deployment approach.
- (c) Outcome: reduced decision delays when AI aligns with hospital workflows.
- (d) Limitation: limited generalisability, supports Mercer’s need for localised validation.

[8] Eriten S. Artificial intelligence-assisted triage in the emergency department: Current status and future perspectives. *Turk J Emerg Med.* 2024;24(1):12-25. (Anchor paper)

- (a) Problem: human variance in manual triage decisions.
- (b) Method: a narrative review of AI-triage credibility.
- (c) Outcome: confirms AI can standardise sorting.
- (d) Limitation: model fragility when biometric fields are missing, the exact technical gap this pilot addresses.

[9] Fenn A, et al. Real-time ML models for ED flow optimization. *J Health Inform.* 2025;12(4):210-225.

- (a) Problem: prolonged ED wait times.
- (b) Method: real-time machine-learning deployment.
- (c) Outcome: optimised patient flow.
- (d) Limitation: tested only under perfect network stability, validates our offline-first architectural constraint.

[10] Harrison C, Patel V. Implementation of machine learning in emergency departments: A systematic review. *J Emerg Med.* 2025;15(2):44-58.

- (a) Problem: technically sound models failing upon clinical deployment.
- (b) Method: a systematic review of implementation failures.
- (c) Outcome: workflow-integration constraints cause roughly 70% of failures.
- (d) Limitation: review-level evidence rather than a single deployment case, still justifies our stakeholder analysis and Friction Protocol.

[11] Lewis M, Dubois A. AI emergency triage: Promise, proof gaps, and security-governance risks in real-world deployments. *Cureus.* 2025;17(1):e0512. (Course case study) **[Ethics / Safety / Governance]**

- (a) Problem: security and governance risks of deploying AI triage in live settings.
- (b) Method: a policy analysis of real-world deployments.
- (c) Outcome: identified severe gaps in data-privacy frameworks.

(d) Limitation: largely theoretical focus, nonetheless mandates our adherence to Grenada data law and WHO principles.

[12] O'Connor S, Trent K. Designing a Human-in-the-Loop AI System for Severity-Aware Triage. Int ISCRAM Conf Proc. 2026. **[Ethics / Safety / Governance]**

(a) Problem: medical-liability and safety risks of fully autonomous triage AI.

(b) Method: comparative testing of human-in-the-loop architectures.

(c) Outcome: human-in-the-loop systems markedly reduced medical-liability exposure.

(d) Limitation: conference-proceedings evidence rather than a peer-reviewed journal, still validates Mercer's non-autonomous mandate.

[13] Okoro J, Silva M. Algorithmic aversion vs. alert fatigue: A comparative ethnographic study. J Emerg Nurs. 2025;51(2):201-215. (Course case study) **[Ethics / Safety / Governance]**

(a) Problem: why senior clinical staff disable AI alerts.

(b) Method: ethnographic shadowing of nursing staff.

(c) Outcome: found a 60% system-bypass rate for interruptive pop-ups.

(d) Limitation: single-site ethnographic sample, directly informs our Passive-Vigilance UI mitigation.

[14] Picard J. Measuring inter-rater variation and triage-level disagreement. Ann Emerg Med. 2023;82(4):411-423.

(a) Problem: manual acuity-grading variance among triage nurses.

(b) Method: a retrospective inter-rater agreement analysis.

(c) Outcome: found severe inter-rater disagreement linked to cognitive fatigue.

(d) Limitation: NHS-based cohort, limited generalisability to Caribbean settings, hence our local validation plan.

[15] Tyler R, Chang H, Mensah J. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A global scoping review. Acad Emerg Med. 2024;31(3):245-258.

(a) Problem: fragmented evidence on global ML triage architectures.

(b) Method: a global scoping review.

(c) Outcome: models are technically credible but under-validated prospectively.

(d) Limitation: scoping-review breadth over depth - supports our decision to run a 12-week localised simulated deployment.

[16] World Health Organization. Ethics and Governance of Artificial Intelligence for Health. Geneva: WHO; 2021. **[Ethics / Safety / Governance]**

(a) Problem: absence of a shared global ethical framework for health-AI deployment.

(b) Method: an expert consensus process producing six governing principles (autonomy, well-being, transparency, accountability, equity, sustainability).

(c) Outcome: provides the baseline governance framework this pilot maps every design decision to.

(d) Limitation: principles are deliberately jurisdiction-agnostic and non-binding, hence our supplementary reliance on the Grenada Data Protection Act (2023).

9. References

Full citations for every source referenced across this proposal, in alphabetical order:

- Agius M, et al. Two-stage XGBoost model for triage prioritisation in high-volume emergency services. *J Emerg Med*. 2025;49(2):112-124.
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- World Health Organization. Ethics and Governance of Artificial Intelligence for Health. Geneva: WHO; 2021.